

CS341 DataLab AI Exp

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Experiment Y

In ChatGPT 5, I prompted with the skeleton file for `bits.c` and my initial Design Document without the examples I worked out by hand. The prompt was "Using the skeleton file `bits.c`, write code to implement the methods described in the `Design.pdf` "

The code from this prompts is in `bitsY.c`

GPT did well with the easier problems like `howManBits` and `True3/4` and `True5/8`. The code Failed the other tests and didnt code within the standards defined at the top of the document. The Score for the `./btest` was 12/24 and that inst considering the operations threshold we had to be under.

Overall I think my designs were good and since I worked them out by hand with simple examples, they work. GPT just has some minor holes in the logic it codes with.

Experiment Z

Once in a while, I asked GPT5 to design a pseudocode outline for the 6 problems. This time I gave it the skeleton code and the Section 4 in the PDF so it has the rules. I also made sure to ask it to write the design in pseudocode since that is how I did my original design before Experiment Y, just so things are in the same format.

The output for the plaintext pseudocode is in Canvas under **CS341 ExpZ GPT Designs.pdf**.

The code is in `bitsZ.c`

GPT Did the same with my design and its design when it comes to the score, still 12/24. There were more casting errors and errors relating to the coding style we had to use. When it comes to the operations, GPT did worse with its own design.